Machine Learning Essentials

Core ideas, terminology, model evaluation, and embeddings explained for practical AI development.

What machine learning means

Machine learning (ML) is the practice of building systems that learn patterns from examples rather than relying only on hand-written rules. A model is a function fitted to data that predicts a target or represents patterns in new observations. The exact performance depends on data quality, the learning method, and evaluation design.

Three learning paradigms

- Supervised learning: training examples include a target label. Classification predicts a category (spam vs. not spam); regression predicts a number (such as a delivery time).
- Unsupervised learning: examples do not have target labels. Clustering groups similar observations; dimensionality reduction compresses features while retaining useful structure.
- Reinforcement learning: an agent chooses actions and receives feedback as rewards, learning a policy for sequential decisions. It is different from ordinary classification.

Features, labels, and data splits

A feature is an input variable; a label is the target for a supervised prediction. A common workflow separates data into a training set to fit the model, a validation set for development decisions, and a test set reserved for the final unbiased check. Exact split ratios are contextual, not fixed rules.

Data leakage happens when training uses information that would not be available at prediction time or inadvertently includes details from the test set. Leakage can make an evaluation look better than real-world performance.

Classification evaluation metrics

A true positive (TP) is a correctly predicted positive; a false positive (FP) is an incorrectly predicted positive; a false negative (FN) is a missed positive.

- Precision = TP / (TP + FP). Of everything predicted positive, how much was actually positive?
- Recall = TP / (TP + FN). Of all actual positives, how many did the model find?
- F1 score = 2 × precision × recall / (precision + recall) when the denominator is nonzero. F1 balances precision and recall.

Accuracy is the share of all predictions that are correct. On strongly imbalanced datasets, accuracy alone may hide poor detection of the minority class. For regression, mean absolute error (MAE) measures the average magnitude of prediction errors.

Overfitting, underfitting, and generalization

Overfitting means a model captures training-specific noise and performs worse on new data. Underfitting means a model is too simple, not adequately learning the signal even in training data. Generalization describes performance on previously unseen examples.

Common countermeasures against overfitting include gathering representative data, simplifying a model, regularization, early stopping, and careful validation. A model should be compared against a basic baseline before drawing strong conclusions.

What embeddings do

An embedding maps a text fragment, image, or another object into a numeric vector. Similar meaning can correspond to nearby vectors depending on the model and its training. Text embeddings are used in semantic search because related text can be found even when it does not contain identical keywords.

An embedding vector is not a reliable factual statement by itself. A nearest-neighbor result is related in the embedding space, but it may still be wrong, incomplete, or irrelevant to the user's precise information need.

Practical connection to RAG

Retrieval-augmented generation uses an embedding model to represent user queries and document chunks, a retrieval method to select likely relevant content, and a language model to generate an answer from retrieved context. Evaluate retrieval quality separately from answer quality.

Questions this document can answer

Q: What is the difference between precision and recall?
A: Precision checks the correctness of predicted positives; recall checks how many real positives were found.

Q: What does overfitting mean?
A: The model has learned training-specific patterns that do not generalize well to new examples.

Q: Why are embeddings useful for semantic search?
A: They let a system compare numeric representations of text and find related meaning without requiring identical wording.